

RTL-SDR Receiver — Cheat Sheet

Chain K — Hardware & Systems Foundations · one-page recall, not first-time learning

SDR Fundamentals

Term	Meaning
IQ sample	Complex number ($I + jQ$) — magnitude AND phase, at one instant
AM demod	magnitude = $\sqrt{I^2 + Q^2}$
FM demod	rate of phase change between consecutive samples
Correct chipset	RTL2832U + R820T2 — buy one explicitly sold as "RTL-SDR"

FM/AM & GQRX

Symptom	Fix
Weak/noisy at every frequency	USB port RF noise — use a USB extension cable
Broadcast FM mode	WFM (mono/stereo) in GQRX
Waterfall display	Frequency (x) vs time (y), color = signal strength

ADS-B Decoding

Field	How to extract
Downlink Format (DF)	Top 5 bits of byte 0 — DF 17 = ADS-B Extended Squitter
ICAO address	Bytes 2-4 (hex chars 2-7) — fixed per airframe, never changes
Decoder	<code>dump1090</code>
Example	<code>8D4840D6202CC371C32CE0576098</code> → DF 17, ICAO 4840D6

APT Weather Sats & Antennas

Fact	Detail
APT frequency	~137MHz
Best antenna	QFH (quadrifilar helix) — frequency + circular-polarization matched
Biggest single lever	Antenna choice/placement — not software settings
Common failure	Whip antenna, no ground plane, indoors — often unusable